

# RAJESWARI T

Chennai, India | +91 91506 00890 | pthangarajaraji@gmail.com  
linkedin.com/in/rajeswari-t-427132322 | github.com/Rajeswari-2802 | rajeswari-2802.github.io

## CAREER OBJECTIVE

---

Electronics and Communication Engineering student aiming for a career in Embedded Systems and VLSI. Hands-on experience across STM32, ESP32-S3 and Raspberry Pi platforms, sensor interfacing and embedded C, with project work spanning industrial condition monitoring, safety systems and vehicle-to-vehicle communication. Looking to apply these skills to real engineering products while growing in an industry setting.

## EDUCATION

---

|                                                  |             |
|--------------------------------------------------|-------------|
| <b>Saveetha Engineering College</b> , Chennai    | CGPA: 8.05  |
| B.E. Electronics and Communication Engineering   | 2023 – 2027 |
| <b>Chennai Government Model School</b> , Chennai | 72.6%       |
| Higher Secondary (12th)                          | 2022 – 2023 |

## PROJECTS

---

### **Vector Aid – V2V Smart Ambulance Alert System** (*Team Project; Final Year Project*)

- Built a vehicle-to-vehicle RF alert system on two Raspberry Pi 4 Model B nodes that warns nearby vehicles of an approaching ambulance.
- Ambulance node fuses NEO-M8N GPS position and MPU6050 heading with OpenRouteService routing, and broadcasts it over an nRF24L01 link to receiver units.
- Tools:** Raspberry Pi 4 Model B, nRF24L01, NEO-M8N GPS, MPU6050, Python, OpenRouteService API

### **SUMA-TEX – Smart Universal Maintenance Assistant for Textile MSMEs**

- Developed a multimodal condition-monitoring system for textile power-looms using ESP32-S3 and multiple sensors for predictive maintenance.
- Fuses vibration, acoustic, temperature and motor-current signals, with a FastAPI backend and ML-based fault classification.
- Tools:** ESP32-S3, MPU6050, MAX9814, DS18B20, SCT-013, Python, FastAPI, AI/ML

### **Advanced Gas Detection and Emergency Shutdown System** (*Top 5, Mini Project Competition*)

- Developed an Arduino-based gas leakage detection and emergency shutdown system with real-time alerts and servo-based valve control.
- Tools:** Arduino Uno, MQ-6, LCD, Servo, Buzzer, Embedded C

### **Industrial Machine Fault Monitoring System using STM32**

- Developed a real-time machine monitoring system using sensor data acquisition through the STM32 ADC for basic fault detection.
- Tools:** STM32, Sound Sensor, Temperature Sensor, ADC, STM32CubeIDE

## INTERSHIPS

---

### **NSIC – Industrial Based Embedded Systems with AI & IoT**

- Hands-on experience with microcontrollers, sensors, embedded programming and IoT-based applications.

### **NSIC – Development of PLC Controls and Industrial Automation**

- Gained practical experience in PLC programming, industrial automation and control systems.

## TECHNICAL SKILLS

---

**Microcontrollers / Boards:** STM32, ESP32-S3, Raspberry Pi 4, Arduino Uno, 8051

**Programming:** C, Embedded C, Python

**Tools:** STM32CubeIDE, Keil, Arduino IDE, Vivado, Cadence, LTSpice, Proteus

**Hardware:** ADC, GPIO, Sensor Interfacing, RF Communication (nRF24L01), Circuit Design

**Soft Skills:** Critical Thinking & Troubleshooting, Collaborative Teamwork, Time Management, Problem Solving

## CERTIFICATIONS

---

NIELIT: Embedded for Beginners • NIELIT: VLSI for Beginners • NPTEL: Introduction to IoT • NPTEL: Modern Digital Communication Techniques • IBM Z: Datathon Hackathon • I4C: DEEPTech Hackathon

## ACHIEVEMENTS

---

- Mini Project “Advanced Gas Detection and Emergency Shutdown System” ranked Top 5 in a Mini Project Competition.